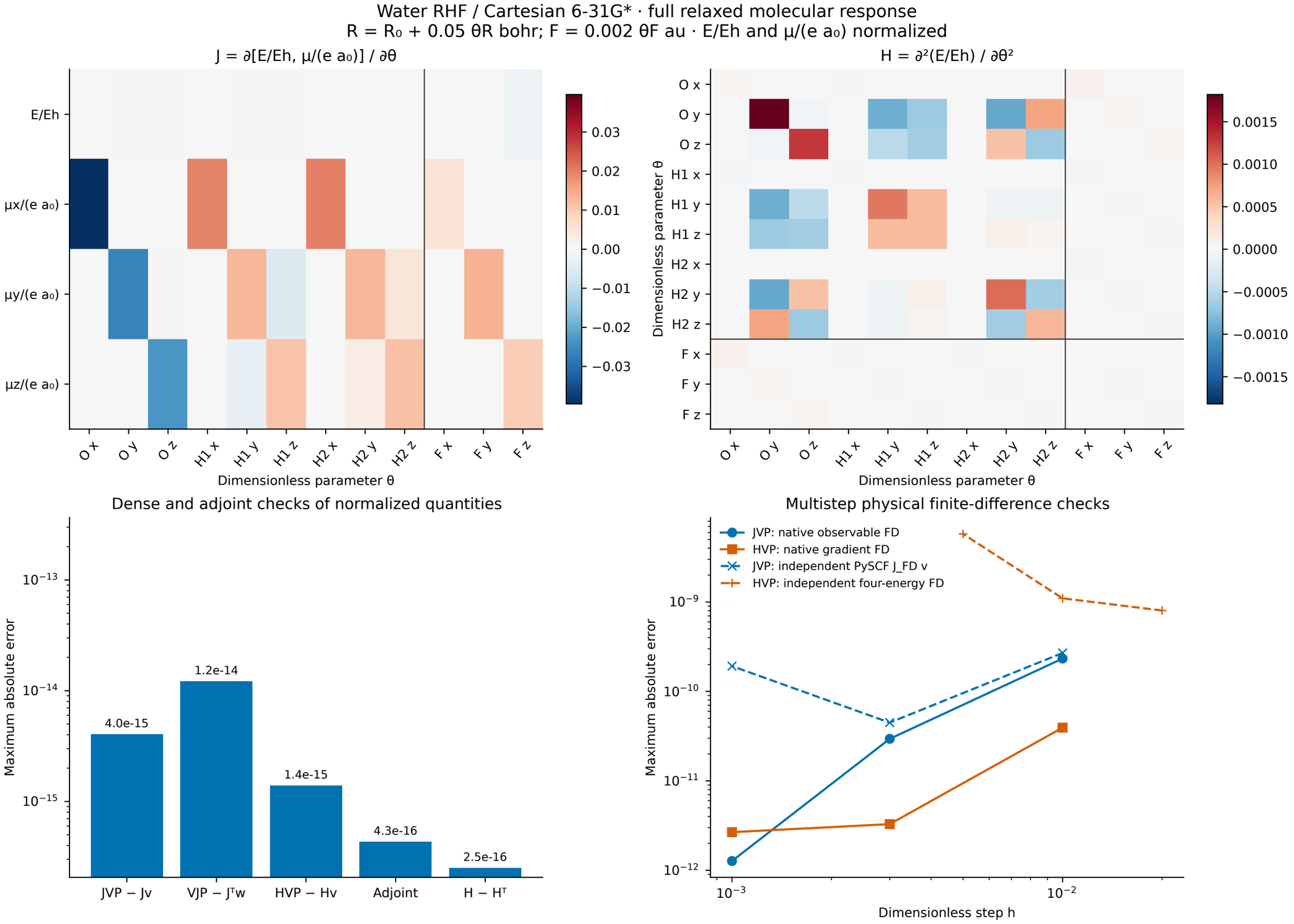
Water RHF / Cartesian 6-31G* · full relaxed molecular response
R = R₀ + 0.05 θR bohr; F = 0.002 θF au · E/Eh and μ/(e a₀) normalized
J = ∂[E/Eh, μ/(e a₀)] / ∂θ
E/Eh
μx/(e a₀)
μy/(e a₀)
μz/(e a₀)
O x
O y
O z
H1 x
H1 y
H1 z
H2 x
H2 y
H2 z
F x
F y
F z
Dimensionless parameter θ
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
H = ∂²(E/Eh) / ∂θ²
Dimensionless parameter θ
0.0015
0.0010
0.0005
0.0000
−0.0005
−0.0010
−0.0015
Dense and adjoint checks of normalized quantities
Maximum absolute error
4.0e-15
1.2e-14
1.4e-15
4.3e-16
2.5e-16
JVP − Jv
VJP − JᵀW
HVP − Hv
Adjoint
H − Hᵀ
Multistep physical finite-difference checks
JVP: native observable FD
HVP: native gradient FD
JVP: independent PySCF J_FD v
HVP: independent four-energy FD
Maximum absolute error
Dimensionless step h